SAINT MICHAEL COLLEGE OF CARAGA

THE IMPACT OF 2021 MODERNIZATION PROGRAMS AS PERCEIVED BY THE BFP PERSONNEL

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CHAPTER 1

THE PROBLEM AND ITS SCOPE

Background of the study

During emergencies, fire protection seeks to protect people, property, and the environment. The Bureau of Fire Protection (BFP) is responsible for this endeavor in the Philippines. Rescue operations, fire prevention, and incident response are among the agency's primary responsibilities. It also looks into fire incidents and enforces fire safety rules. It also works with volunteer fire brigades to strengthen coordination and improve community preparedness.

However, the BFP continues to experience several long-standing challenges. Many stations still use old equipment, lack enough personnel, and operate with limited resources. The agency's capacity to respond promptly and provide uniform fire protection nationwide is frequently hampered by these issues. The government implemented modernization measures to improve the BFP's overall performance and operations and address these problems. The BFP Modernization Act of 2021 focuses on three main goals. First, it upgrades fire stations and other facilities. Second, it improves emergency response systems. Third, it strengthens coordination between BFP staff and volunteer firefighters. Evaluating the program is important to see if it works effectively and can be maintained over time (MJBAS, 2022).

International experience demonstrates that modernization efforts in fire services significantly improve emergency response performance. In the United States, Japan, and Australia, studies show that holding regular training sessions and improving communication systems helps lower fire-related deaths and limit property damage (Smith & Jones, 2020; Kim et al., 2021).

In the United Kingdom, firefighters now use modern equipment along with digital command systems. This combination improved station coordination. Fire stations responded to emergencies faster and coordinated more effectively as a result (Johnson, 2019).

Similarly, in Singapore, modernization efforts have emphasized interoperability between government fire services and volunteer brigades, resulting in a more unified and efficient response system. These international experiences provide valuable insights into the potential outcomes of modernization initiatives in developing nations.

In the Philippines, fire service modernization became more critical after Republic Act No. 11589, or the BFP Modernization Act of 2021. This law calls for building new fire stations, upgrading facilities, and buying modern fire trucks and equipment. Local studies indicate these programs have improved operations in some areas, but challenges persist. Training, maintenance, and the equitable distribution of resources between urban and rural stations still require improvement. Volunteer fire brigades, who serve as essential partners, also report mixed levels of support. While some have access to shared resources, others still lack adequate assistance (Dela Cruz, 2022; Reyes, 2023).

Although the modernization program aims to make the BFP a world-class fire service, little research has examined how staff and volunteers feel about the changes. This study addresses that gap. It examines whether modernization meets operational needs, improves skills, and strengthens cooperation between government and volunteer teams. It looks at equipment, training, organizational efficiency, and morale. By identifying both strengths and weaknesses, the study provides practical guidance for better planning and resource management.

This research is helpful for many groups. Policymakers and government agencies can assess the effectiveness of modernization investments and identify areas for improvement. The BFP can use the results to improve training, maintain equipment, and optimize operations. Finally,

the study adds to public safety research and can guide future fire service programs. Overall, the goal is to strengthen fire protection, make communities safer, and increase public trust in emergency services.

Review of Related Literature

Before this study, the chapter reviews existing literature. It looks at articles, books, and dissertations. The goal is to understand the topic and guide the research.

Relate Literature

The BFP Modernization Act of 2021 (Republic Act No. 11589) aimed to strengthen the Bureau of Fire Protection. It upgraded facilities, staff, and equipment. It also established specialized services, including emergency medical response and HAZMAT operations. The BFP modernization program includes fire protection services, force restructuring, organizational development, capacity building, technology upgrades, specialized services, and human resources development (Lawphil, 2021).

Santos et al. (2024) reported improvements in equipment, technology, training, and facilities nationwide. Modern equipment helps staff respond faster. Training improves skills and job satisfaction. Upgraded facilities and communication systems boost morale, safety, and community involvement (Department of Budget and Management, 2023; Cruz et al., 2023).

These areas are key parts of the modernization program. Legislative documents and recent studies confirm their importance.

The ten-year program aims to make the BFP more effective and keep communities safer. Scholars say modernization is not only about equipment. It also includes training, morale, and efficiency (Delarosa, 2021). Organizational reforms increase staff competence and motivation when they fit field needs (Alonzo, 2019).

The program also values fire volunteers. Volunteers help the BFP reach more communities. Training them and including them in local programs improves disaster preparedness and public trust (Santos, 2020).

Studies in other countries show the same. Volunteers included in training feel more involved and contribute more to safety (Martinez, 2018). These show that both staff and volunteers are important for success. Some studies warn that modernization may face challenges. Bureaucracy, uneven resources, and volunteers paying for fuel and equipment reduce effectiveness (Reyes, 2022).

Age

The age profile of BFP personnel helps us understand how the 2021 Modernization Program affects them. Most BFP personnel in the Philippines are young, mostly aged 21 to 40 (Bautista & Rodriguez, 2022).

In Laguna province, almost one-third (29.79%) of personnel aged 21 to 30 made up one group. Another large group included those aged 31 to 35. These indicate that all staff members are in the early to mid-career stage. Younger personnel adapt more easily to new technologies and updated procedures (EPRA International Journal of Multidisciplinary Research, 2023).

In Sorsogon province, most personnel fell within the 25-34 age range. In the National Capital Region, many were 30 to 34. This mix of individual insight and experience helps balance adaptability with knowledge. Most staff in these age groups are male, reflecting traditional roles in firefighting. Female participation is growing. The age profile shows why training and capacity-building programs should focus on staff who can benefit most from modernization (All Multidisciplinary Journal, 2023).

Heraldero & Lomarda (2021) studied BFP personnel in Bukidnon. They found that staff aged 21 to 35 were more engaged in training programs, like the Fire Basic Recruit Course and leadership seminars. Younger personnel were mainly male and new to the service. They should

be ready to use new procedures and technologies. Continuous training improves fire prevention and suppression performance.

Looking at these age groups helps assess the program's effectiveness and sustainability. A younger workforce can adopt innovations, follow new procedures, and maintain the gains from modernization. This profile fits the national goal of improving public safety through technology and training. Age remains an important factor for planning future programs in the BFP (Scribd Firefighter Profile, 2025).

Gender

Understanding the gender profile of respondents, especially BFP personnel and volunteer firefighters, is important for understanding how the 2021 Modernization Program affects people. Studies consistently show that the workforce is primarily male, but female participation is increasing (Civil Service Commission, 2022).

A study in Sorsogon Province found that about 70% of BFP personnel were male, while females accounted for about 30% of the workforce. These show that firefighting has been male-dominated, but agencies are making efforts to include more women. Female personnel hold various ranks and contribute significantly to firefighting, emergency medical services, and disaster management (Multidisciplinary International Journal of Research and Development, 2024).

The modernization program highlights gender-specific roles and supports female personnel. For example, specialized training and leadership programs help the careers of female fire officers take on greater responsibilities and leadership positions. These initiatives aim to remove gender barriers and give women equal opportunities for career growth. This approach aligns with national goals for gender equality and the integration of women into traditionally male

fields such as firefighting and emergency response (Multidisciplinary International Journal of Research and Development, 2024).

A study in Dagupan City shows that male personnel still dominate operational roles, and female participation is limited. Gender balance varies across locations, showing the need for local strategies to include more women. Recruitment, training, and awareness programs can help ensure FEMA personnel are more fairly represented and involved in modernization efforts across the board (EPRA International Journal of Multidisciplinary Research, 2023).

In Cavite, research shows that although firefighting is still predominantly male, the number of female firefighters has increased, especially in administrative, training, and support roles. This change reflects a shift in organizational norms encouraged by the modernization program, which promotes gender equality. Inclusion policies and gender sensitivity training are important parts of this transformation to modernize the BFP's workforce culture (Scribd, 2025).

In South Korea, female firefighters play strong leadership roles in community programs and fire prevention education (Kim, 2021). These studies suggest that gender balance not only supports equality but also improves efficiency and service delivery, which the BFP is gradually adopting as part of its modernization agenda.

In the Philippines, several studies show how gender dynamics in the BFP are changing. Reyes and Dela Cruz (2022) found that the number of women in technical and administrative roles increased after modernization introduced gender-sensitive training and fair promotion systems. Morales (2021) noted that gender inclusivity in recruitment improves public trust, mainly when female firefighters serve as approachable role models. These findings underscore the role of gender equality in enhancing professionalism and fostering greater public engagement within the BFP.

Rank

According to Sanidad-Leones et al. (2023), their study examined perceptions of the impacts of BFP modernization in the Caraga Region and found that rank and role influenced views on improvements in fire protection services, organizational development, and training effectiveness. Higher ranks perceived greater strategic benefits, while frontline personnel emphasized operational improvements. They also highlighted the importance of continuous training and equipment upgrades.

Organizational-change research suggests that perceptions of such reforms often vary by rank or hierarchical level. For example, Jones et al. (2018) reported that executives maintained a generally positive outlook during a hospital restructuring. In contrast, supervisory and rank-and-file staff referred more to conflict and divisions and expressed more negative attitudes toward the change than did executives.

Likewise, Wynen et. al (2022) report that employees at higher (strategic) levels tend to view organizational reforms more favorably than lower-level or technical staff. These studies indicate that higher-ranking officers often appreciate the goals of modernization more than those closer to frontline duties.

By contrast, mid-level supervisors and ordinary personnel are more likely to focus on the challenges or uncertainties introduced by large-scale changes. Sunada et al. (2025) evaluated the competencies of BFP personnel in the National Capital Region and found that modernization has improved the capabilities of fire officers across ranks. They emphasized that training programs enhanced personnel skills in firefighting, rescue, and fire prevention, leading to improved performance aligned with rank responsibilities.

Moreover, Kim et al. (2025) examined Korean firefighters' views during a conversion to the national civil service. They found that overall satisfaction with the reform increased with rank

within the fire organization. Lower-ranking roles (such as paramedics in their study) reported lower satisfaction compared to line firefighters. In the context of modernization, rank-and-file FP personnel, or those in lower-tier roles, may be less optimistic than senior officers. These findings underscore that rank and position can shape how employees experience and evaluate organizational change: upper echelons often have a broader perspective on objectives, while lower ranks may be more sensitive to operational impacts.

Additionally, Van den Ende et al. (2020) studied Dutch fire service regionalization and found "notable tensions" between professional managers and volunteer firefighters during the reform. Their analysis highlights a clash between the drive for professionalization and volunteers' local, intrinsic motivations. In practical terms, volunteer BFP firefighters (who are numerous in the Philippines) may have distinct concerns or sense-making about modernization, for instance, questioning whether new systems adequately support their role or fearing loss of traditional autonomy.

Furthermore, Ramos et al. (2025) highlighted that personnel in higher ranks possess advanced decision-making and leadership skills, enabling them to implement modernization initiatives effectively. In contrast, lower-ranking personnel demonstrate strong operational skills but may require additional training to align with modernization objectives fully.

Similarly, Lomarda (2025) found that age and experience—demographic factors that often correlate with rank—influence the performance and perceptions of BFP personnel, with younger personnel demonstrating greater effectiveness in fire suppression and prevention tasks. Longa (2024) emphasized that tailored training programs should consider rank and length of service, as personnel at different ranks exhibit varying capabilities and experiences that can impact the success of modernization efforts.

Moreover, Sabalosa (2024) noted that higher-ranking personnel, such as station chiefs, face unique challenges and responsibilities that provide them with distinct insights into the program's implementation, reinforcing the importance of rank in evaluating the modernization program's overall effectiveness. Collectively, these studies suggest that rank is a critical factor in understanding how BFP personnel and fire volunteers perceive the modernization program, and addressing rank-specific needs can enhance the program's impact and sustainability.

Years of service

The study by Lomarda (2025) shows that years of service affect how BFP personnel see modernization. Lomarda studied personnel in the Southern Districts of Bukidnon. The study found that younger personnel performed well in fire suppression and prevention. It also found a link between fire skills and overall performance. Longer-serving personnel use their experience to shape their views on reforms. Their practical knowledge and loyalty influence how they respond to changes. Age and sex did not show a direct link to performance. However, skills and experience from longer service strongly affect how personnel perceive modernization.

Newer personnel with fewer years of service often see modernization differently. They enter the BFP with higher expectations. They adapt more easily to new technologies. De Guzman (2022) found that shorter-service personnel accept modernization programs more readily because they are less tied to traditional practices. This generational difference creates a potential gap in acceptance and implementation. The BFP needs policies that include both older and newer personnel.

In the context of the BFP Modernization Act of 2021 (R.A. 11589), which provides a ten-year modernization plan, years of service are important for evaluation. Lopez (2022) notes that modernization should not be measured only by technology. It should also consider how personnel

at different service levels respond. Including both new and seasoned staff in decision-making ensures modernization meets practical needs and plans for the future, years of service shape how personnel judge modernization programs. Senior staff rely on their experience in the BFP or volunteer unit. They compare modernization efforts to past limitations in equipment, training, and systems.

Cruz (2020) reports that long-serving personnel use outdated systems as a benchmark. They either support modernization if they see improvements or criticize it if changes seem impractical. Research shows that tenure in public service builds professional identity and culture (Miller et al., 2017). This culture shapes how personnel interpret modernization. Long-serving staff often value upgrades such as advanced fire trucks, communication systems, and emergency medical equipment. They have seen the system evolve from limited resources to modern tools (Perez et al., 2021).

Newer personnel often have different views. They adapt to technology more quickly. Delos Santos (2022) confirms that shorter-service personnel are more open to modernization because older practices have less influence on them. This difference highlights the need for inclusive policies. Years of service also affect the adaptation to training. Hernandez (2020) found that senior personnel may find digital-based modernization programs harder to learn. Training that considers different experience levels reduces resistance. It helps modernization succeed.

Finally, the BFP Modernization Act of 2021 sets a ten-year roadmap. Years of service remain a key factor in evaluating success. Lopez 2022) stresses that modernization should balance technology with staff perspectives. Including both senior and new personnel ensures practical and forward-looking results.

Fire Protection Services

Fire protection services are organized systems to prevent, respond to, and reduce fires and related emergencies. The International Fire Service Training Association (FSTA, 2020) defines fire protection as encompassing education, prevention, preparedness, and suppression. In the Philippines, the Bureau of Fire Protection (BFP) under Republic Act No. 6975 leads these services.

The enactment of the BFP Modernization Act of 2021 (R.A. 11589) aims to professionalize the agency and equip it with modern technologies to address evolving fire and emergency challenges. The law mandates the establishment of fire protection services and rescue hotlines in all local government units, prioritizing municipalities without fire services (Lawphil, 2021).

Before modernization, BFP encountered significant operational constraints. Equipment was outdated, staffing levels were inadequate, and resources were insufficient. The Commission on Audit (2018) reported delays in fire station construction and fire truck delivery, noting that only 263 of the 945 planned stations had been completed by mid-2018.

Additionally, Ayalin (2019) reported that 308 municipalities lacked fire stations, while 59 had no fire trucks. The Philippine Star (2018) highlighted a shortage of over 26,000 personnel and insufficient protective equipment. These findings were further confirmed in the BFP 2018 Accomplishment Report, which documented project delays and suspensions.

As well as, Resource distribution was uneven, favoring urban areas over rural municipalities. Poor facility design, weak maintenance systems, and inadequate communication infrastructure reduced response effectiveness in provincial areas (DOF-BFP Report, 2019). Firefighters faced difficulties accessing remote communities, increasing casualties. De Leon (2020) emphasized gaps in advanced training, while Villanueva (2020) noted low public awareness due to limited outreach in informal settlements.

Moreover, modernization has led to measurable improvements in fire suppression and rescue operations. The integration of the national 911 system, GIS technology, digital monitoring platforms, and drones has enhanced deployment efficiency and reduced response times (Delos Santos, 2022).

Also, the BFP Modernization Act of 2021 provides legal and financial mechanisms for acquiring modern equipment, constructing new fire stations, and upgrading nationwide communication systems. It also emphasizes recruitment, training, and professional development of personnel and volunteers (BFP, 2021).

The BFP safeguards lives and property through fire suppression, prevention, investigation, and emergency medical response (Delos Santos, 2021), in accordance with the Fire Code of the Philippines (R.A. 9514). Reyes (2022) highlighted the BFP's role in inspections, permit issuance, and public education. Navarro (2020) emphasized the agency's expanded function in disaster and rescue operations.

Furthermore, Fire prevention initiatives are central to risk reduction. Gonzalez (2022) reported that programs such as Oplan Ligtas na Pamayanan significantly reduce structural fires. Villanueva (2021) noted improved fire investigation practices, while Martinez (2022) and Tan (2022) documented enhancements in equipment, communication, and inter-agency coordination.

Fire protection is vital to community safety and economic stability. The World Fire Statistics Centre (2021) found that strong fire services reduce casualties and property losses. DILG (2023) emphasized that effective fire services support national Disaster Risk Reduction and Management (DRRM) goals.

In addition, modern fire protection prioritizes prevention through education, school campaigns, and community drills, which reduce risks before emergencies occur (Haddow et al.,

2021). Cote and Bugbee (2019) and Stull (2015) stressed that trained and well-equipped fire departments improve public trust and reduce economic losses.

Globally, modernization enhances fire service performance and disaster resilience. Singapore employs smart fire stations and digital dispatch systems (Tanaka, 2021). Japan uses GIS and predictive analysis (Kobayashi, 2020). The United States applies standardized command systems and advanced training (US Fire Administration, 2020), while the United Kingdom utilizes digital fire modeling (Brown, 2021). Australia integrates climate-based forecasting to manage wildfires (Peters, 2022).

The US Fire Administration (2025) emphasized that continuous modernization, advanced training, and public education significantly reduce casualties and economic losses.

Force Restructuring and Organizational Development

Force restructuring and organizational development refer to deliberate changes in an agency's internal structure, leadership, and management systems to improve coordination, accountability, and efficiency. Specifically, these reforms include reorganizing departments, decentralizing authority, creating specialized divisions, and establishing new positions. According to Burke (2020), organizational development consists of planned actions that enhance institutional capacity through continuous improvement.

Moreover, Burke (2022) explained that restructuring improves organizational effectiveness through leadership development, personnel training, and structural redesign. In the context of public safety institutions such as the Bureau of Fire Protection (BFP), restructuring aligns personnel deployment and hierarchy with operational demands, thereby enabling agencies to respond more efficiently to emergencies.

In line with this, the BFP Modernization Program aims to establish a compact, efficient, and modern firefighting force in compliance with R.A. 9514. Under its implementing rules, the BFP is required to submit a revised organizational structure and staffing pattern to the Department of the Interior and Local Government (DILG) and the Department of Budget and Management (DBM) within 90 days. Additionally, the law mandates a qualification program for officers and personnel in accordance with Civil Service Commission guidelines (Lawphil, 2021).

Furthermore, R.A. 11589 reorganizes the BFP by creating additional positions, adjusting ranks, and establishing specialized units to address modern firefighting and emergency response requirements (Congress of the Philippines, 2021).

Prior to modernization, the BFP faced internal structural weaknesses. For instance, Lantaco (2019) reported that an outdated organizational structure limited local decision-making. Similarly, Garcia (2020) found staffing imbalances, with urban areas adequately staffed while rural stations lacked personnel and specialized units. In addition, Mendoza (2018) observed unclear promotion pathways that reduced morale.

Likewise, Villanueva (2019) identified weak coordination between administrative and operational units, leading to inefficiencies. At the same time, Lopez (2020) highlighted insufficient leadership training and the lack of continuous professional development, which reduced organizational adaptability.

Following modernization, reforms resulted in notable organizational improvements. For example, Rivera (2022) reported that the restructuring of regional and command offices improved authority distribution and accelerated decision-making. Moreover, the creation of specialized divisions such as Strategic Management and Fire and Emergency Medical Services enhanced integration between administrative, operational, and technical functions.

Consistent with these findings, Indama (2023) noted clearer hierarchies and the introduction of performance-based leadership systems. As a result, these reforms addressed prior inefficiencies and increased organizational adaptability.

At the national level, agencies supported restructuring through capacity-building initiatives. Specifically, Soriano (2023) reported that the DILG and Civil Service Commission implemented leadership and competency programs for BFP officers. In addition, Lim (2022) noted that partnerships with universities enabled personnel to pursue education in public safety management and leadership.

Supporting this perspective, Burke (2017) emphasized that organizational redesign paired with leadership development improves effectiveness. Similarly, North (2022) stated that clear structures enhance accountability, reduce redundancy, and speed decision-making. Collectively, these reforms prepare the BFP to meet increasing fire safety demands, particularly in urbanized areas.

Empirical evidence further supports this outcome. Heladero et al. (2025) examined BFP personnel in Southern Bukidnon and found high effectiveness in fire suppression operations. However, they recommended increasing staffing levels, providing continuous training, defining clearer roles, and implementing competency profiling and career laddering to further strengthen professionalism.

Beyond the national context, international experiences offer benchmarks for organizational reform. For instance, Anderson (2021) reported that Australia's Integrated Command and Control System improved coordination and resource allocation. Likewise, Yamamoto (2022) found that Japan's collaborative disaster management models strengthened inter-agency coordination.

Finally, the International Association of Fire Chiefs (2019) stated that restructuring enhances coordination, accountability, and decision-making. In support of this, Kim et al. (2020) added that leadership development combined with restructuring improves morale and professionalism. Meanwhile, Williams (2022) noted that the United Kingdom's National Fire Chiefs Council adopted performance-based management systems, which influenced Philippine reforms. Similarly, Johnson (2023) reported that the United States Fire Administration strengthened leadership structures and task forces to address emerging hazards.

Capability, Material, and Technology Development

The BFP Modernization Program under R.A. 11589 aims to transform the agency into a capable, efficient, and modern fire service organization. Specifically, the program prioritizes the upgrading of firefighting equipment, vehicles, and protective gear, while simultaneously phasing out outdated and unserviceable resources (Lawphil, 2021). Through this approach, the legal framework establishes capability, material, and technology development as core components of modernization.

In support of this framework, Grant (2019) emphasized that capability development enhances skills, knowledge, and competence in high-risk professions. Similarly, Cote and Bugbee (2019) noted that upgrading fire trucks, protective equipment, and rescue tools improves firefighter safety and response effectiveness. Along the same lines, Haddow et al. (2017) stated that modern technologies such as GIS, digital command systems, and real-time communication enhance coordination and situational awareness.

Prior to modernization, multiple studies documented serious deficiencies in BFP resources. For instance, the Commission on Audit (2017) reported that many fire trucks were old or unserviceable, with some dating back to the 1950s, and that at least 349 municipalities lacked

fire trucks. In addition, the BFP Logistical Report (2021) showed that 197 municipalities still lacked fire stations and trucks, with BARMM having 63 towns without fire trucks.

Moreover, the Commission on Audit reported that by mid-2018, only 621 of the planned 1,057 fire trucks had been procured and only 263 of 945 targeted fire stations were completed. Contributing to this issue, Lopez (2020) cited inadequate maintenance budgets, while Castillo (2020) identified supply-chain issues and lack of technical expertise as barriers to adopting new tools. Taken together, these findings reveal uneven resource distribution and structural limitations in sustaining operational readiness.

Beyond equipment constraints, training deficiencies further limited BFP operational capability. Santos (2020) and Delos Reyes (2019) reported the absence of structured training programs in advanced rescue operations, chemical fire management, and digital systems. As a result, many BFP units continued to rely on manual processes, indicating slow technological adaptation.

To address these gaps, R.A. 11589 allocates funds for upgrading vehicles, tools, and communication systems while phasing out obsolete equipment (Congress of the Philippines, 2021; BFP, 2021). Consistent with this direction, Castillo (2022) noted that ICT systems now support dispatch and monitoring, signaling a shift toward digital integration.

Following implementation, recent studies indicate improvements in BFP capability. For example, the DILG (2023) confirmed the construction of new fire stations, installation of command systems, and implementation of real-time tracking tools nationwide, supported by partnerships with private and international organizations. At the same time, the BFP Training Institute now leads capacity-building initiatives focusing on digital literacy, equipment handling, and international standards (Echavaria, 2022).

Furthermore, the modernization program mandates the use of advanced firefighting tools, including aerial ladders, rescue vehicles, communication systems, and breathing apparatuses that meet international standards (BFP, 2021). In relation to this, Villanueva (2022) observed that rank-based training strengthened competencies in fire suppression, emergency response, and disaster management. Likewise, Lorenzo (2022) highlighted the use of drones and GIS mapping for improved situational awareness.

At the national level, DOST (2022) supports the development of local firefighting tools and detection systems, while regional training centers expand access to capacity-building programs (De Guzman, 2021). Meanwhile, international partners such as JICA and NFPA provide technical support, and the DILG (2022) introduced an Integrated Information System for real-time reporting and resource monitoring.

Globally, studies reinforce the importance of continuous innovation in fire services. For instance, USFA (2019) emphasized simulation-based training and leadership development. Similarly, the UK Fire and Rescue Service standardize firefighter competencies nationwide (Home Office, 2020), while IFSTA (2020) promotes virtual reality and online simulations for training.

In the Asian context, Tanaka (2021) highlighted the use of robotics and artificial intelligence in Tokyo, whereas Kim (2021) and Kobayashi (2020) reported the use of drones, IoT, and thermal imaging in South Korea and Japan. At the same time, Australia and Canada invest in simulation centers and data analytics (Harris, 2020), while FEMA (2021) funds advanced tools such as GIS, sensors, and drones. Overall, these international practices emphasize sustained investment in technology to enhance readiness and public safety.

Specialized Services Development

Specialized Services Development focuses on strengthening advanced response teams that manage high-risk and complex emergencies. In this context, these services require specialized training, equipment, and coordinated operations (Harison, 2018). Under the BFP Modernization Program, specialized units include Hazardous Materials (HAZMAT), Urban Search and Rescue (USAR), Emergency Medical Services (EMS), and Fire and Arson Investigation Units (BFP, 2022). Moreover, the BFP Modernization Act of 2021 mandates the creation of units for high-rise fires, forest fires, aircraft incidents, chemical emergencies, and medical response (Lawphil, 2021).

In relation to these mandates, Castro (2022) emphasized that these units enhance the BFP's capacity to respond to industrial explosions, structural collapses, and large-scale disasters. Similarly, Santos and Mendoza (2023) noted that EMS divisions equipped with modern ambulances and trained paramedics now deliver pre-hospital medical care as part of fire operations. In addition, Ogle et al. (2021) highlighted the establishment of the Fire and Arson Investigation Division (FAID), which strengthened investigative accuracy through scientific methods and inter-agency coordination. Taken together, these developments indicate a shift toward a multi-skilled and integrated emergency response agency.

Before 2021, the BFP experienced significant limitations in delivering specialized services. For example, Baluran (2018) reported insufficient funding, training, and equipment for HAZMAT, USAR, and EMS operations, resulting in delayed and inefficient responses. Likewise, Navarro (2019) noted that outdated rescue tools forced firefighters to rely on manual methods, increasing operational risk.

Furthermore, the Commission on Audit (2018) identified incomplete equipment procurement, delayed projects, and persistent gear shortages, which prevented many stations from forming fully functional rescue or HAZMAT teams. At the same time, García (2019) reported

unclear operational guidelines that hindered coordination with partner agencies. In addition, Lopez (2020) found the absence of standardized protocols and certifications, leading to inconsistent performance across regions.

Beyond operational issues, institutional constraints further limited specialized services prior to modernization. Specifically, Manalo (2018) reported limited collaboration with international fire organizations, restricting access to global best practices. Meanwhile, Villanueva (2020) observed that bureaucratic processes and budget constraints delayed equipment acquisition. Similarly, Reyes and Alonzo (2020) noted the lack of training facilities and qualified instructors, which restricted skill development.

Therefore, Morales (2021) warned that these structural and institutional gaps increased risks to both responders and the public. Collectively, these findings underscore the urgency for comprehensive modernization to address systemic weaknesses.

Following policy reforms, the implementation of Specialized Services Development adopted a phased and strategic approach. For instance, Perez (2022) reported that specialized units were strategically deployed in high-risk areas to improve response times. Likewise, Garcia (2023) observed that newly established regional training centers now provide continuous education and certification for specialized responders.

In support of this approach, technology plays a crucial role in implementation. Accordingly, Nakamura (2022) emphasized that thermal imaging cameras, GIS-based fire mapping, and digital communication systems enhance operational efficiency. At the same time, Alvarez (2023) noted that partnerships with local governments and private organizations improved logistics, funding, and coordination.

In addition to technical upgrades, Lopez and Santiago (2023) stressed the importance of physical and mental health support programs for firefighters, recognizing the demands of

specialized emergency work. Altogether, these measures demonstrate that specialized services modernization has transitioned from policy to operational practice.

From an international perspective, literature confirms the importance of specialized services in modern fire management. For example, Reuters (2025) reported that Greece strengthened specialized firefighting units to improve wildfire response. Similarly, the United Nations Development Programme (UNDP) has supported countries in developing specialized fire units through modern equipment, training, and planning, paralleling Philippine efforts.

Consistent with global trends, Thompson (2022) stated that the Philippine strategy aligns with best practices observed in Singapore and Tokyo. In the same vein, Luga et al. (2023) noted that adopting Incident Management Systems and standardized protocols brings the BFP closer to International Fire Service Accreditation Congress (IFSAC) standards.

At the national level, R.A. 11589 formally institutionalized USAR, HAZMAT, EMS, and Fire and Arson Investigation units (BFP, 2022). Accordingly, Rodriguez (2022) stated that the law ensured legal authority and funding support for these services. Furthermore, Santos and Velasco (2023) reported that partnerships with JICA and USAID enhanced technical skills through specialized training and field exercises.

Operationally, Martinez (2022) highlighted the deployment of mobile command centers, rescue boats, and drones to improve situational awareness. In addition, Villanueva (2023) and Lopez (2022) documented specialized programs for high-angle, confined-space, and water rescues. Meanwhile, the Office of Civil Defense (OCD, 2022) reported improved inter-agency coordination, while the BFP adopted NFPA (2020) guidelines to professionalize specialized responders. Finally, Villamor and Cruz (2023) emphasized that sustained funding supports continuous advanced training.

Human Resources Development

The BFP Modernization Program emphasizes the development of a comprehensive training system for both personnel and volunteers, including mandatory entry-level training and advanced facilities (Lawphil, 2021). Within this framework, Human Resource Development (HRD) is central to the initiative, as it aims to improve skills, professionalism, and operational efficiency among public safety personnel. Accordingly, Swanson and Holton (2019) defined HRD as the systematic development of human potential through training, career growth, and organizational improvement in order to achieve individual and institutional goals.

In the same vein, Werner and DeSimone (2012) emphasized that in government modernization, continuous HR training is essential for adapting to new technologies, management systems, and policy reforms. Taken together, these perspectives establish HRD as a strategic foundation of the BFP modernization agenda.

Consistent with this direction, the BFP Modernization Act of 2021 (R.A. 11589) formally institutionalized HRD as a pillar of operational strengthening. Specifically, the law established the BFP Training Institute, professional education programs, and leadership development initiatives to support workforce competence and professionalism.

Prior to 2021, HRD implementation faced serious constraints. For example, Fernandez (2018) observed inconsistent training delivery across regions. Similarly, Domingo and Perez (2019) reported outdated training materials and inadequate facilities, which limited career progression and emergency preparedness. Collectively, these findings highlight the need for a standardized and legally supported HRD system.

Moreover, several studies documented HRD challenges before modernization. Aguilar (2020) found that limited funding and training opportunities resulted in delayed promotions, low motivation, and poor personnel retention. In addition, Castillo (2019) noted unclear career

pathways, which weakened leadership development. Likewise, Reynoso (2020) highlighted weak coordination with other agencies, limiting opportunities for inter-agency training.

At the same time, external pressures compounded these issues. For instance, Cruz (2020) noted that emerging hazards and frequent fire disasters demanded skills many personnel lacked. Furthermore, Espinoza (2021) identified the absence of stress management training, leading to burnout and reduced morale. Meanwhile, Villareal (2020) emphasized that gender inclusivity in training was often neglected, restricting opportunities for women in the fire service.

Following the enactment of R.A. 11589, the implementation of HRD reforms significantly transformed the BFP. As a result, the BFP Training Institute now offers structured programs covering technical, managerial, and leadership competencies (BFP, 2021). Additionally, digital learning platforms and virtual simulations have modernized instruction and expanded access (Cruz, 2023).

In support of these reforms, Reyes and Bautista (2023) documented collaborations with the Civil Service Commission, DILG, and universities to deliver specialized leadership programs. At the same time, the Department of Science and Technology (DOST, 2022) contributed by integrating fire safety technologies into training curricula. Moreover, Fernandez (2021) noted the inclusion of welfare programs such as mental health counseling, insurance, and family support.

In terms of outcomes, Garcia (2023) observed that digital performance monitoring systems now promote transparency, accountability, and consistency in training and evaluation. Similarly, Villanueva (2023) emphasized that sustainability depends on adapting global best practices to local conditions.

Globally, HRD is recognized as a strategic investment in fire services. For example, Harris (2020) reported that Australia's Country Fire Authority implements annual competency checks, leadership workshops, and mental health programs. Likewise, Canada's Fire Service College

incorporates diversity, wellness, and psychological readiness into training. In the same context, Japan's Fire and Disaster Management Agency applies tiered technical, leadership, and psychological training for lifelong learning (Kitagawa, 2020).

In the United States, HRD is standardized through national institutions. Specifically, Hulett (2018) explained that the US Fire Administration and National Fire Academy provide leadership, incident management, and technical rescue training. Furthermore, FEMA (2019) supports certification systems such as the Pro Board to ensure consistent competency across departments. Together, these practices demonstrate that structured HRD enhances professionalism and reduces training disparities, principles reflected in BFP modernization.

At the local level, HRD reforms increasingly reflect international standards. For instance, Dela Peña (2022) reported that the BFP Training Institute now offers competency-based programs in Fire Investigation, EMS, and USAR. In addition, Bradley (2021) found that leadership immersion and merit-based promotion improved morale and career satisfaction.

Similarly, Garcia and Mendoza (2023) noted that HRD promotes continuous education and technical certifications such as HAZMAT and Fire Safety Inspection Training. Consequently, Lopez (2022) emphasized that these initiatives build a workforce capable of handling complex emergencies. Meanwhile, the Civil Service Commission (2022) highlighted the role of partnerships in strengthening leadership, accountability, and ethics.

Finally, Cruz (2023) and Villareal and Santos (2023) observed that digital learning and simulation-based training now align BFP HRD with global fire education standards. In addition, Bautista (2022) added that values formation and professionalism are embedded in HRD programs to sustain service excellence.

Theoretical Framework

This study draws on key ideas from organizational change, technology adoption, and employee motivation. These theories help explain how the 2021 modernization programs may be affecting the Bureau of Fire Protection (BFP) and the volunteer firefighters who support it. By looking at these perspectives, we can better understand how modernization influences not only the tools and processes but also the people who make the fire service work.

This study is anchored on the Modernization Theory developed by Walt Whitman Rostow (1960). Rostow's theory explains how organizations and societies progress through stages of development from being traditional and less efficient to becoming modern, dynamic, and highly capable. He believed that modernization occurs when institutions adopt new technologies, improve their systems, and enhance their personnel, resulting in greater productivity and progress.

In relation to this study, Rostow's Modernization Theory helps explain how the 2021 Bureau of Fire Protection (BFP) Modernization Program aims to transform the agency into a more advanced and efficient organization. The program's components, such as upgrading fire equipment, improving facilities, providing continuous training, and developing specialized services, reflect the process of modernization that Rostow described. These efforts signify the BFP's movement toward a more professional and well-equipped institution capable of delivering better public service.

The relevance of this theory lies in showing how modernization can influence the performance and perception of BFP personnel. The independent variable, which is the 2021 Modernization Program, includes improvements in fire protection services, force restructuring and organizational development, capability, material and technology development, specialized services development, and human resource development. The dependent variable, which is the

perception of BFP personnel, reflects how these modernization efforts are experienced by those working within the organization.

Applying Rostow's Modernization Theory suggests that as the BFP continues to modernize through better tools, training, and systems, its personnel are likely to feel more motivated, capable, and confident in performing their duties. In this way, modernization becomes not just an institutional goal but a direct factor in improving employee morale and service quality.

Overall, Rostow's Modernization Theory serves as the guiding framework for this study. It supports the idea that modernization programs, like that of the BFP, play a vital role in driving institutional progress and shaping how personnel perceive and experience change within their organization

Schematic Diagram

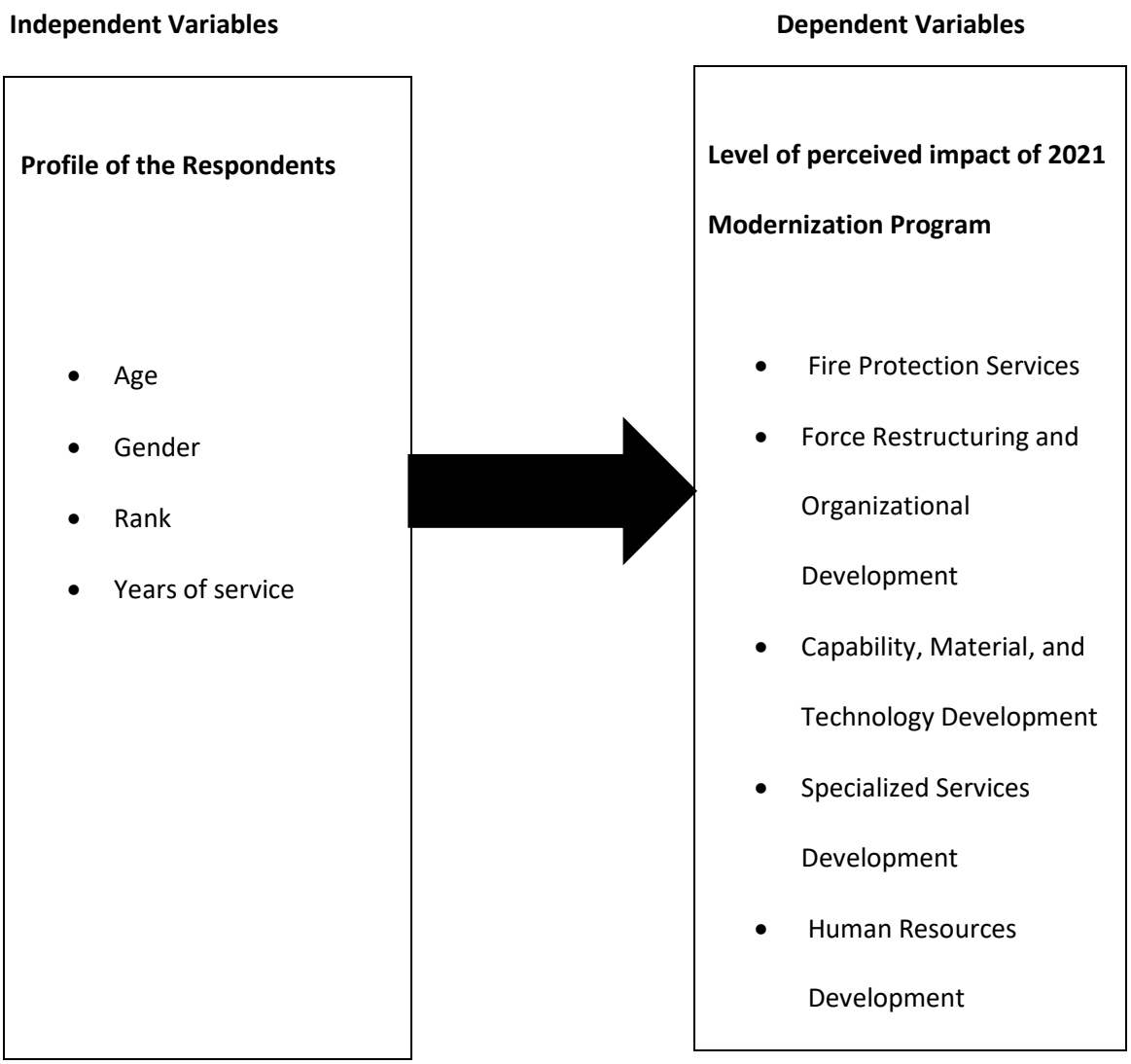


Figure 1. The Schematic Diagram shows the relationship between the dependent and independent variables of the study.

Statement of the Problem

This study aims to determine the perceived impact of the 2021 Modernization Program among BFP personnel, considering their demographic profiles. Specifically, it seeks to answer the following questions:

1. What is the profile of the respondents in terms of:

   1.1 age;

   1.2 gender;

   1.3 rank; and

   1.4 years of service?

2. What is the level of perceived impact of the respondents on the 2021 Modernization Program in terms of:

   2.1 Fire Protection Services;

   2.2 Force Restructuring and Organizational Development;

   2.3 Capability, Material, and Technology Development;

   2.4 Specialized Services Development; and

   2.5 Human Resources Development?

3. Is there a significant relationship on the perceived impact of the 2021 Modernization Program when grouped according to their profile?

Hypothesis

H0: There is no significant relationship in the perceived impact of the 2021 Modernization Program when grouped according to profile.

Definition of Terms

The following terms are defined both operationally and conceptually for the purposes of this study:

Age. It refers to how long a particular respondent has lived.

Gender. Refers to identifying a person as male, female, or other, thereby helping recognize gender differences.

Rank. Refers to the person who holds a position or job within the Bureau of Fire Protection, or as a volunteer, such as a firefighter, officer, or team leader, which might affect how they experience the modernization.

Years of Service. Refers to how long a person has worked or volunteered with the firm, giving insight into whether more experienced members view changes differently from newer ones.

Fire Protection Services. Refers to enhancing fire prevention, suppression, and emergency response capabilities through better coverage and the establishment of rescue hotlines in all local government units in Buenavista, Nasipit, and Carmen.

Force Restructuring and Organizational Development. Refers to the reorganization of the BFP's structure and workforce to improve operational efficiency and align with modernization goals through optimized command, roles, and internal processes.

Human Resources Development. Refers to the continuous training, professional development, and capacity building of BFP personnel and volunteers to equip them with advanced skills and knowledge necessary for effective fire and emergency operations.

Capability, Material, and Technology Development. Refers to the acquisition and deployment of advanced firefighting equipment, technology, and materials to enhance the operational effectiveness and technological edge of the BFP.

Specialized Services Development. Refers to the enhancement of specialized firefighting and emergency services, including chemical, biological, radiological, nuclear, and explosive emergency response capabilities, supported by expert services and advanced coordination systems.

CHAPTER 2

RESEARCH METHODOLOGY

This chapter covers the methods used for data collection and analysis in the present study. It explains the various stages of the research, which include the research design, research materials and methods, research environment, population and sample, research data and analysis plan, and research ethical standards.

Research Design

This study employed a descriptive–correlational quantitative research design. The descriptive part focused on identifying the demographic profile and perceptions of Bureau of Fire Protection (BFP) personnel regarding the 2021 Modernization Program. The correlational part aimed to determine whether there is a significant relationship between the perceived implementation of the modernization program and the perceived impact on fire protection services. A correlational design is appropriate when the researcher seeks to determine the degree of relationship between two or more variables without manipulating them. This design allows the study to identify the association between modernization and its perceived effects on BFP operations while maintaining objectivity using numerical data (Creswell, 2018).

To represent the respondents, the researchers used purposive sampling. This sampling approach ensures that the researchers intentionally select participants based on their specific characteristics or experience that are relevant to the research question. By using this method, the researchers obtained more accurate and insightful information, enabling them to draw stronger and more meaningful conclusions about the impact of modernization on fire protection services.

Research Materials and Methods

This study used a quantitative approach. A survey questionnaire served as the main instrument for data collection. The researchers divided the questionnaire into two main sections. The first section collected information on respondents' demographic profiles, including age, gender, rank or volunteer role, and years of service.

The second section focused on their perceptions of the BFP Modernization Program, covering five key areas: Fire Protection Services; Force Restructuring and Organizational Development; Capacity, Material, and Technology Development; Specialized Services Development; and Human Resource Development. The researchers applied a four-point Likert scale, with four indicating Strongly Agree, three indicating Agree, two indicating Disagree, and one indicating Strongly Disagree. This approach helped the researchers better understand how BFP personnel and fire volunteers view the organization's modernization efforts.

Research Environment, Population, and Sample

Research Environment

The researchers conducted the study in Buenavista, Nasipit, Carmen, and Butuan City in Agusan del Norte, Philippines. The study was conducted specifically in Buenavista, Nasipit, Carmen, and Butuan City. The location is densely populated, featuring a mix of residential, commercial, and industrial areas. It has a diverse population, including BFP personnel. The researchers chose Buenavista, Nasipit, Carmen, and Butuan City to collect responses, allowing them to capture diverse perspectives on the BFP modernization program, making these locations ideal for the study.

Population and Sample

Table 1. Respondents of the study

Respondents	Population	
	F	**%**
BFP NASIPIT	15	13.2
BFP CARMEN	13	11.4
BFP BUENAVISTA	20	17.5
BFP LIBERTAD	11	9.7
BFP BUTUAN CITY	47	41.2
BFP AMPAYON	8	7.0
Total	**114**	**100.0**

Table 1 shows the population of the BFP NASIPIT, BFP CARMEN, BFP BUENAVISTA, BFP LIBERTAD, BFP BUTUAN CITY, AND BFP AMPAYON.

Research Data and Analysis Plan

The researchers first secured approval from the research adviser and the dean of the CCJE Department. After the approval, we sent a Letter of Request to the Municipal Fire Marshals of Buenavista, Nasipit, and Carmen, seeking permission to obtain the total number of BFP personnel and fire volunteers for the survey. The Municipal Fire Marshals approved the request, allowing the researchers to proceed with the study. Next, we distributed the questionnaires face-to-face to the selected respondents in the three municipalities. After collecting, the research team reviewed each questionnaire for completeness and accuracy, returning any incomplete forms for clarification where necessary. Finally, we will consolidate and process the data. Responses will be checked, tallied, tabulated, and analyzed using frequency and percentage, as outlined in SOP1. In SOP2, we will use the mean, and in SOP3, we will use the pearson correlation coefficient.

Research Ethical Standards

A. Protection of Intellectual Rights

The researchers ensured they respected others' intellectual property by properly citing all sources used in the study. Ideas, concepts, and materials from other authors were acknowledged to uphold honesty and fairness in academic work.

B. Informed Consent

Before participating in the study, the researchers fully informed the respondents about its purpose, goals, and procedures. The researcher explained the participants' roles, described how they would collect and store the data, and clarified how they would use the information. The study included only participants who gave informed consent.

C. Data Privacy and Confidentiality

Protecting respondents' privacy was a top priority. The researchers kept the information from the questionnaires strictly confidential and used it only for research purposes.

D. Voluntary Participation and Freedom to Withdraw

Participation in the study was entirely voluntary. Respondents could refuse to take part or withdraw at any time without facing any penalties.

E. Minimization of Harm and Risk Management

The researchers ensured that no participant experienced any form of physical, emotional, or psychological harm during the study. The researchers made every effort to maintain their comfort, safety, and well-being throughout the process.

F. Beneficence and Contribution to Knowledge

This research aimed to generate valuable insights to improve fire protection services and modernization programs. The study also hoped to contribute to the broader field of public safety and community development.

G. Justice and Fair Participant Selection

Participants were chosen fairly and based on their relevance to the study. No one was excluded or discriminated against. Each respondent received an equal opportunity to participate in the research.

H. Data Integrity and Accuracy

The researchers handled all gathered data carefully and reported the results truthfully. The researchers ensured that the results presented were accurate, reliable, and free of manipulation.

I. Transparency and Honesty in Reporting

The researchers were transparent about all procedures, results, and limitations of the study. The researchers openly acknowledged any possible bias or error to maintain the study's credibility.

J. Use of Patented or Copyright Materials

All copyrighted or patented materials, such as books, questionnaires, and digital tools, were cited correctly and used only with permission. The researchers always respected intellectual property rights.

K. Ethical Considerations for Animal and Human Traits

Since the study involved human participants, their rights, dignity, and welfare were always respected. The researchers tolerated no form of exploitation or harm throughout the research.

L. Responsible Use of AI and Other Related Technologies

The researchers used artificial intelligence and digital tools to process data or prepare content. They applied these technologies responsibly and ethically to preserve the study's originality and integrity.

M. Ethical Clearance and Institutional Approval

Before the study began, the researchers secured the necessary approvals and clearances from the appropriate authorities. These ensured that all research activities followed institutional and academic ethical standards

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ORGANIZATION	POSITIONS HELD
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ORGANIZATION	POSITIONS HELD

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Membership/Affiliations

ORGANIZATION	POSITIONS HELD

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Membership/Affiliations

ORGANIZATION	POSITIONS HELD